ECE 6410 HW04

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Question 1

Question Statement

FM signal generation. Reproduce the single-tone FM signal shown in Figure P1.

Given: $\{f_1 = 1 \text{ kHz}, f_c = 20 \text{ kHz}, \beta = 5\}$ The number of samples, ns=256 for this example.

Generated Plot

Figures 1 and 2 show the reproduced FM signal provided in the homework.

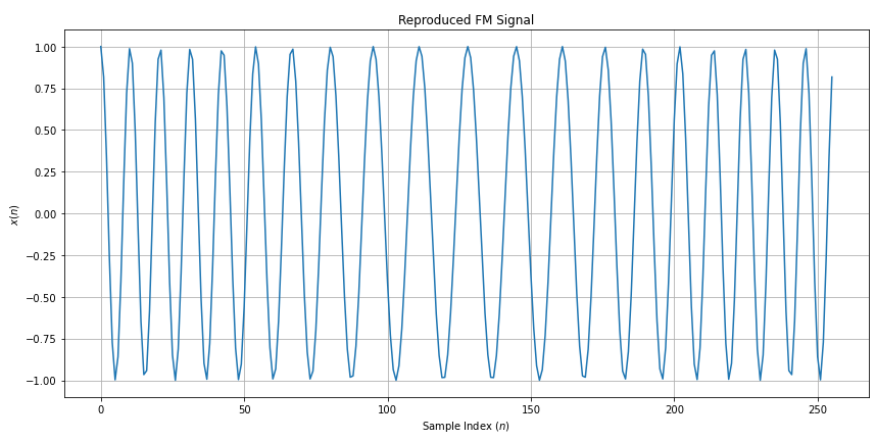

Figure 1: Time domain of reproduced FM signal.

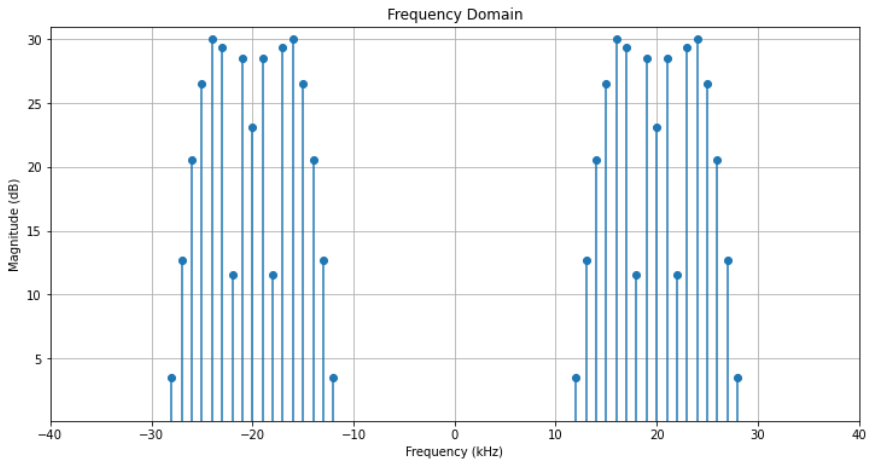

Figure 2: Frequency domain of reproduced FM signal.

Code

Below is the code used to generate the plots shown above.

```python

def p1():
    # constants
    f1 = 1e3          # hertz
    fc = 20e3         # hertz
    B = 5             # beta
    ns = 256          # samples
    A = 1             # amplitude (set to 1 for now)

    # fs = n * f1
    fs = 100e3
    fs = ns *f1       # sampling frequency
    n = np.arange(ns)
    t = n / fs

    # calculate omega values
    w1 = 2*np.pi*f1
    wLO = 2*np.pi*fc

    # plug into given equation
    x_n = A * np.cos(wLO * t + B * np.sin(w1 * t))
    #x_n = A * np.cos(2 * np.pi * fc * t + B * np.sin(2 * np.pi * f1 * t))

    # plot
    plt.figure(figsize=(12, 6))
    plt.plot(n, x_n)
    plt.title(r'Reproduced FM Signal')
    plt.xlabel('Sample Index ($n$)')
    plt.ylabel('$x(n)$')
    plt.grid(True)
    plt.tight_layout()
    plt.show()

```

```python
    # Frequency domain

    # calc FFT
    X_f = np.fft.fft(x_n)
    freqs = np.fft.fftfreq(ns, 1/fs)

    X_mag = np.abs(X_f)

    # dB conversion
    X_db = 20 * np.log10(X_mag/20)
    # normalize to 30 dB as shown in HW
    X_db_norm = X_db - np.max(X_db) + 30

    # plotting
    plt.figure(figsize=(12, 6))
    plt.stem(freqs / 1000, X_db_norm)
    plt.title(f'Frequency Domain')
    plt.xlabel('Frequency (kHz)')
    plt.ylabel('Magnitude (dB)')
    plt.grid(True)
    plt.xlim(-40, 40)
    plt.ylim(0.1, 31)
    plt.show()
```

Question 2

Question Statement

Refer to Figure P1. A Z_0=50Ω cable communication system operates at f=900 MHz. Two matched line amplifiers are placed between cables runs with lengths shown. The RF source has an output power of P_s=13dB_m.

a) Set up a link budget using RG-85 cable. Determine the signal power level at the input of the receiver, P_{rdB}.

b) If the equivalent noise power at the input of the receiver is N_R=15dB_μ, determine SNR_{dB}.

c) If the minimum receiver SNR=+10dB, would switching to RG8X coaxial cable be a solution? Show your work.

d) Suppose access specifications required RG-174 to be used (narrow dimaeter and bending radius). What minimum gain for the pair of line amplifiers would you recommend?

View the attached document for answers and work.

Question 3

Question Statement

The consumer-band (L1) frequency for GPS is f_{L1}=1.575 GHz. From the handout, the GPS constellation is medium Earth orbit (MEO) with an altitude of approximately R_{min}=20,200km. Assume the staellite EIRPdB = PTdB + GTdB $\approx$ 26dB. The noise power at the receiver is N=8 dB_f.

a) Sketch the satellite link budget. Assume R_{min} is directly overhead the receiver. How would you estimate R_{max}? Show youre work with a sketch and an estimate of the earth's radius (given below). You may assume a perfect sphere and neglect the equatorial 'bulge'.

b) If the satellite P_T is 2W what is the Gain of the transmit atenna?

c) Use R_{min} to calculate a minimum path loss.

d) Estimate the received power, P_{rdB}.

e) What is the estimated SNR at the receiver?

View the attached document for answers and work.

Question 4

Question Statement

The DoD military band GPS operates at f_{L2}=1.227GHz with and $EIRP_{LP}$ ≈22dBW.

a) Use R_{min} to calculate a minimum path loss.

b) Estimate the received power.

c) What is the estimated SNR at the receiver?

View the attached document for answers and work.